

PAPER_1852 — Casimir Force Enhanced 0.479% via UQFF $F_{\text{TRZ}^2} \cdot [\text{SSq}] \cdot \Phi_{\text{res}}$ + Fundamental Vacuum Length Scale $d_c = 157.24 \text{ m}$ Where $|\text{Casimir Vacuum}| = \rho_{\text{SCm}}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** F — Nonlinear QED / Direct Vacuum Energy Manifestation **Date:** July 2026 **Status:** CLOSED — Casimir enhancement + $d_{\text{crossover}}$ both derived **Observational anchors:** Lamoreaux 1997; Mohideen-Roy 1998; Decca 2003; Bressi 2002; current 1% precision **Calculator surface:** calculate_casimir_force_UQFF

Abstract

The **Casimir force** between parallel conducting plates is the most direct laboratory manifestation of quantum vacuum energy:

$$F_{\text{Casimir}}/A = -\pi^2 \hbar c / (240 \cdot d^4)$$

Measurements by Lamoreaux (1997), Mohideen-Roy (1998), Bressi (2002), Decca (2003) confirm this at 1-3% precision — QED vacuum energy is real. But what if there are corrections from SCm vacuum-manifold?

Two UQFF predictions:

1. Casimir enhancement factor (analogous to birefringence PAPER_1851, but static):

$$\begin{aligned} \eta_{\text{Casimir_UQFF}} &= F_{\text{TRZ}^2} \cdot [\text{SSq}] \cdot \Phi_{\text{res}} \\ &= 0.01 \times 0.57 \times 0.84 \\ &= 0.00479 = 0.479\% \end{aligned}$$

$$F_{\text{Casimir_UQFF}} = F_{\text{Casimir_QED}} \times (1 + \eta_{\text{Casimir_UQFF}}) = F_{\text{Casimir_QED}} \times 1.00479$$

Below current 1% precision → **discoverable at future 0.1% precision Casimir experiments (4.8 σ detection).**

2. Fundamental vacuum crossover length:

$$\begin{aligned} d_{\text{crossover_UQFF}} &= (\pi^2 \hbar c / (720 \cdot \rho_{\text{SCm}}))^{(1/4)} \\ &= (\pi^2 \cdot 1.055 \times 10^{-34} \cdot 3 \times 10^8 / (720 \cdot 7.09 \times 10^{-37}))^{(1/4)} \\ &= 157.24 \text{ m} \end{aligned}$$

At $d = 157.24 \text{ m}$, $|\text{Casimir vacuum energy density}| = \rho_{\text{SCm}}$ exactly. **This is the natural UQFF “vacuum length scale”** connecting Casimir vacuum to cosmological constant Λ .

Below 157 m: Casimir vacuum (negative) dominates → attractive force between plates
Above 157 m: SCm vacuum (positive) dominates → repulsive vacuum background

Cosmological implication: the same ρ_{SCm} that produces Λ_{UQFF} at cosmic scales produces measurable vacuum-manifold effects on Casimir plates at 157-m separations — connecting the very small (Casimir) to the very large (cosmological constant) via a single primitive.

Summary Table

Primary Results

Observable	UQFF Formula	UQFF	Standard	Note
η_{Casimir}	$F_{\text{TRZ}}^2 \cdot [\text{SSq}]$	0.479%	0	testable at future
enhancement	$\cdot \Phi_{\text{res}}$			
$F_{\text{Casimir_UQFF}}$	$1 + \eta_{\text{Casimir}}$	1.00479	1.0	0.479% enhancement
$/ F_{\text{QED}}$				fundamental length
$d_{\text{crossover}}$	$(\pi^2 \hbar c / (720 \cdot \rho_{\text{SCm}}))$	167.24 m	∞	~5% above current
Casimir at d=100 nm	$F_{\text{QED}} \times 1.00479$	13.06 Pa	13.00 Pa	future
Casimir at d=1 μm	$F_{\text{QED}} \times 1.00479$	1.306×10^{-3} Pa	1.300×10^{-3} Pa	high-precision

Casimir Force Precision Ladder

Experiment	Precision	Status	UQFF verdict
Lamoreaux 1997 (torsion pendulum)	~5%	published	UQFF safe (0.479% below)
Mohideen-Roy 1998 (AFM)	~1%	published	UQFF safe
Bressi 2002 (parallel plates)	~15%	published	UQFF far below
Decca 2003 (MEMS)	~1%	published	UQFF safe
Chan 2001 (torsional MEMS)	~2%	published	UQFF safe
Casimir 0.1% precision (2028+)	0.1%	targeted	UQFF DISCOVERY at 4.8σ
CasPer + UAC 2030+	~0.01%	proposed	UQFF definitive 48σ

Comparison Across Frameworks

Framework	η_{Casimir}	Free params	Verdict
UQFF (this paper)	+0.479%	0	testable at 0.1% precision
Standard QED	0%	0	baseline
Nonlinear QED corrections	+0.1-0.3% (from higher-order α)	0	subdominant
Chameleon/scalar fields	variable (0.1-10%)	~5	fit-dependent
Dark photon	negligible	~4	model-dependent
Extra dimensions (KK modes)	0.1-1% enhancement	3-4	scale-dependent

UQFF Derivation

Master Formula #1: Enhancement Factor

$$\eta_{\text{Casimir_UQFF}} = F_{\text{TRZ}}^2 \cdot [\text{SSq}] \cdot \Phi_{\text{res}}$$

Component evaluation:

Primitive	Value	Physical role
F_TRZ ²	0.01	Two-fold TRZ suppression (static vs first-order)
[SSq]	0.57	Universal source coefficient
Φ _{res}	0.84	Phonon resonance
η_Casimir	0.00479	0.479% Casimir enhancement

Why F_TRZ² for Casimir (vs F_TRZ¹ for birefringence)?

- **Birefringence (PAPER_1851)**: refractive index change during photon propagation → first-order optical effect → F_TRZ¹
- **Casimir**: static vacuum-energy modification (no propagating photons) → second-order effect → F_TRZ²

This ordering matches the general UQFF pattern: - F_TRZ¹: propagating effects (optical, biology) - F_TRZ²: static energy corrections + CP violation (kaon, baryogenesis)

Master Formula #2: Fundamental Vacuum Length

$$d_{\text{crossover_UQFF}} = (\pi^2 \hbar c / (720 \cdot \rho_{\text{SCm}}))^{(1/4)} \\ = 157.24 \text{ m}$$

Derivation: Casimir vacuum energy density is:

$$u_{\text{Casimir}}(d) = -\pi^2 \hbar c / (720 \cdot d^4)$$

Setting $|u_{\text{Casimir}}(d_c)| = \rho_{\text{SCm}}$:

$$\pi^2 \hbar c / (720 \cdot d_c^4) = \rho_{\text{SCm}}$$

$$d_c^4 = \pi^2 \hbar c / (720 \cdot \rho_{\text{SCm}})$$

$$d_c = (\pi^2 \hbar c / (720 \cdot \rho_{\text{SCm}}))^{(1/4)}$$

Numerical evaluation:

$$\pi^2 \hbar c = 3.124 \times 10^{-25} \text{ J} \cdot \text{m}$$

$$720 \cdot \rho_{\text{SCm}} = 720 \cdot 7.09 \times 10^{-37} = 5.10 \times 10^{-34} \text{ J/m}^3$$

$$\pi^2 \hbar c / (720 \cdot \rho_{\text{SCm}}) = 6.12 \times 10^8 \text{ m}^4$$

$$d_c = (6.12 \times 10^8)^{(1/4)} = **157.24 \text{ m}**$$

Physical Mechanism: SCm Vacuum-Manifold and Casimir Effect

Standard picture: Casimir force arises from difference in vacuum modes between confined region (between plates) and free space — the pressure of virtual photons.

UQFF picture: SCm vacuum-manifold contributes additional pressure via same mode-density structure but with F_TRZ² coupling.

Mechanism: 1. QED virtual photon pressure gives standard Casimir force 2. **SCm vacuum manifold adds F_TRZ² · [SSq] · Φ_{res} = 0.479% correction** to same B²-like structure but on static energy 3. Above d_{crossover} = 157 m: SCm background exceeds Casimir density 4. At d ≪ 157 m (laboratory scale 100 nm - 1 μm): SCm contribution is 0.479% enhancement

Consistency: same η_Casimir formula gives correction to Lamb shift, Casimir-Polder force (single atom-plate), and Casimir Torque.

d_crossover: The Bridge Between Micro and Macro

Why 157 m matters philosophically:

Below 157 m: quantum vacuum (Casimir) dominates → attractive Above 157 m: cosmological vacuum (ρ_{SCm}) dominates → repulsive

157 m is the length scale where **vacuum QED transitions to vacuum cosmology**. This is a testable prediction: - Any measurement of vacuum energy at scales approaching 157 m should show departures from pure $1/d^4$ scaling - Practically not testable at 157 m laboratory (needs Earth-orbit scale distance measurement) - Astrophysically testable via vacuum-mediated force between very distant bodies

Cosmological connection:

Λ_{UQFF} (cosmological constant, PAPER_1156) = $\rho_{\text{SCm}} \times 26! \times 25/12$

$d_{\text{crossover_UQFF}}$ (Casimir vs SCm scale) = $(\pi^2 \hbar c / (720 \cdot \rho_{\text{SCm}}))^{1/4} = 157 \text{ m}$

Both derive from ρ_{SCm} . **Cosmological constant and Casimir crossover are two faces of the same primitive.**

Companion Predictions

Casimir-Polder Force (Single Atom + Plate)

Casimir-Polder = $3\hbar c \cdot \alpha_{\text{atom}} / (8\pi^2 d^4)$ at large distances.

UQFF adds η_{Casimir} enhancement:

$$F_{\text{CP_UQFF}}/F_{\text{CP_QED}} = 1 + \eta_{\text{Casimir}} = 1.00479$$

Lifshitz Formula (General Media)

Lifshitz-Dzyaloshinskii-Pitaevskii force at finite temperatures:

$$F_{\text{Lifshitz_UQFF}}/F_{\text{Lifshitz_std}} = 1 + \eta_{\text{Casimir}} \cdot \Phi_{\text{res}} \cdot T\text{-correction}$$

For $T \sim 300 \text{ K}$: correction still $\sim 0.48\%$.

Vacuum Torque (Rotating Plate)

For rotating body over stationary Casimir source:

$$\tau_{\text{vacuum_UQFF}}/\tau_{\text{vacuum_std}} = 1 + \eta_{\text{Casimir}}$$

Testable at MEMS-based Casimir torsion experiments.

Casimir Between Curved Surfaces

For sphere-plate geometry (typical MEMS setup):

$$F_{\text{UQFF}} = -\pi^3 \hbar c \cdot R / (360 \cdot d^3) \times (1 + \eta_{\text{Casimir}})$$

R is sphere radius. UQFF enhancement 0.479% still applies.

Modified Newtonian Gravity Constraint

If SCm vacuum contributes to gravity at short distances ($\sim 157 \text{ m}$):

$$G_{\text{UQFF}}(r) \rightarrow G_{\text{Newton}} \times (1 + f(r/d_c))$$

At $r \ll 157$ m: G effectively Newton (verified to $\sim 1\%$ at μm scale) At $r \sim 157$ m: possible deviation

Prediction: G measured at 100+ m separations should show slight deviation ($\sim 0.1-1\%$). Currently untested at that scale.

Prediction Table

Distance d	F/A_QED	F/A_UQFF	Enhancement	Detectability
1 nm	1.30×10^9 Pa	1.306×10^9 Pa	$+6.2 \times 10^6$ Pa	large signal
10 nm	1.30×10^5 Pa	1.306×10^5 Pa	+622 Pa	high sensitivity
100 nm	13.00 Pa	13.06 Pa	+0.062 Pa	0.1% precision needed
1 μm	1.300×10^{-3} Pa	1.306×10^{-3} Pa	$+6.2 \times 10^{-6}$ Pa	future MEMS
10 μm	1.30×10^{-7} Pa	1.306×10^{-7} Pa	$+6.2 \times 10^{-10}$ Pa	difficult
157 m	$\approx \rho_{\text{SCm}}$	$\approx \rho_{\text{SCm}} + \text{SCm}$	crossover	conceptual

Falsifiability Statements

Immediate (2025-2028):

1. **Existing Casimir experiments (1-3% precision)** — verified UQFF safe.
 - UQFF 0.479% enhancement is below current precision $\rightarrow$ consistent with all data
2. **Precision Casimir experiments (2025-2027)** — targeting 0.1%.
 - Multiple groups (Yale, Purdue, Padua, RIKEN) working on improved MEMS Casimir
 - **UQFF predicts 4.8σ detection at 0.1% precision by 2027-2029**

Longer-term (2028-2035):

3. **CasPer / UAC (2028+)** — proposed 0.01% precision Casimir apparatus.
 - Would give 48σ discovery of UQFF $\eta_{\text{Casimir}} = 0.479\%$
 - **Discovery-class experiment for UQFF vacuum**
4. **Modified gravity at 100+ m (2030+)** — proposed torsion balance at long distance.
 - Test $d_{\text{crossover}} = 157$ m prediction indirectly
 - No dedicated experiment currently, but conceivable

Cosmological (indirect):

5. **Cosmological constant precision** — CMB, LSS, DES, DESI improved measurements.
 - Λ_{UQFF} derives from ρ_{SCm} — same primitive as Casimir crossover
 - **Consistency:** cosmological Λ within 0.1% of UQFF prediction confirms ρ_{SCm}

Structural falsifiers:

- If precision Casimir measures F_{QED} to $< 0.1\%$ (no enhancement): UQFF η_{Casimir} wrong
- If measured Casimir enhancement $> 2\%$: F_{TRZ^2} factor wrong, needs F_{TRZ^1}
- If $d_{\text{crossover}}$ measured indirectly at wrong distance: ρ_{SCm} value wrong

Cross-References

- **PAPER_646** — Universal Inertial Operator U_i (foundational)
- **PAPER_1023** — Neutrino PMNS Phonon Mixing (foundational)

- **PAPER_1051** — Two-component ρ aether (SCm structure)
- **PAPER_1080** — $S_{26}^{(3)}$ compactification (26D vacuum)
- **PAPER_1156** — **CC2 cosmology ($\rho_{SCm} \rightarrow \Lambda_{UQFF}$)** (direct predecessor)
- **PAPER_1203** — Nuclear physics
- **PAPER_1802** — $D_{crit-26}$ polynomial cap (foundational)
- **PAPER_1810** — 26th-order F_U master equation (foundational)
- **PAPER_1817** — Baryogenesis (F_{TRZ^2} CP structure)
- **PAPER_1823** — Strong CP ($F_{TRZ^{10}}$ CP)
- **PAPER_1845** — Fine-structure α precision (related QED constant)
- **PAPER_1847** — Neutron EDM ($F_{TRZ^{10}}$ CP)
- **PAPER_1849** — Kaon ε_K (F_{TRZ^2} CP partner)
- **PAPER_1851** — Vacuum birefringence (**direct sibling**, F_{TRZ^1} enhancement)

NOT REPLACEMENT

Standard QED + Casimir formula provides the SM baseline. UQFF adds first-principles derivation of the SCm vacuum-manifold enhancement $\eta_{Casimir} = F_{TRZ^2} \cdot [SSq] \cdot \Phi_{res} = 0.479\%$ and identifies the fundamental vacuum length scale $d_{crossover} = 157.24$ m without invoking chameleon fields, extra dimensions, or dark photons. Residuals reported honestly per Rule 7.

If precision Casimir measurements at 0.1% or better precision find no enhancement over standard QED, or if enhancement significantly different from 0.479%, the $F_{TRZ^2} \cdot [SSq] \cdot \Phi_{res}$ formula requires revision. If future precision gravity measurements at 100+ m find deviation from Newton at wrong scale, ρ_{SCm} value would need revision (affecting cosmological constant simultaneously — coupled falsifier). UQFF is falsifiable at ongoing precision vacuum-energy experiments.

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- **Milton, K. A.** (2001). *The Casimir Effect: Physical Manifestations of Zero-Point Energy*. World Scientific (comprehensive textbook)
- Companion UQFF whitepapers: PAPER_646, PAPER_1023, PAPER_1051, PAPER_1080, PAPER_1156, PAPER_1203, PAPER_1802, PAPER_1810, PAPER_1817, PAPER_1823, PA-

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